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SELF ATTENTION REFERENCE FOR IMPROVED DIFFUSION PERSONALIZATION

Abstract

A method, apparatus, non-transitory computer readable medium, and system for image processing include obtaining a reference image an input prompt describing an image element, identifying an object from the reference image; generating, using an image generation model, image features representing the object based on the reference image, and generating, using the image generation model, a synthetic image depicting the image element and the object based on the input prompt and the image features from the reference image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/558,365, filed on Feb. 27, 2024, in the United States Patent and Trademark Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The following relates generally to image processing, and more specifically to image processing using a machine learning model. Image processing refers to the use of a computer to edit an image using an algorithm or a processing network. In some cases, image processing software can be used for various image processing tasks such as image restoration, image detection, image compositing, image editing, and image generation. For example, image generation includes the use of the machine learning model to generate an image based on an input such as a text prompt or a reference image.

[0003] Image generation refers to generating a synthetic image based on one or more inputs such as a reference image or a text prompt. Image generation is useful in generating new images that have features that are aligned with the inputs. However, conventional models do not generate and integrate specific target details.

SUMMARY

[0004] Aspects of the present disclosure provide a method and system for personalizing an image generation model. According to some aspects, the system receives a text prompt and a reference image depicting an object to generate a synthetic image that depicts the object. In one aspect, an image generation model is used to generate image features including detailed information of the object based on the reference image. The image features are input into one or more attention layers of the U-Net architecture of the image generation model to guide the image generation model to generate the synthetic image.

[0005] A method, apparatus, non-transitory computer readable medium, and system for image processing include obtaining a reference image an input prompt describing an image element; identifying an object from the reference image; generating, using an image generation model, image features representing the object based on the reference image; and generating, using the image generation model, a synthetic image depicting the image element and the object based on the input prompt and the image features from the reference image.

[0006] A method, apparatus, non-transitory computer readable medium, and system for training a machine learning model include obtaining a training set including a reference image depicting an object, an input prompt describing an image element, and a ground-truth image depicting the object and the image element, and training, using the training set, the image generation model to generate image features for the object based on the reference image and to generate a synthetic image depicting the object based on the input prompt and the image features.

[0007] An apparatus and system for image processing include at least one processor, at least one memory storing instructions executable by the at least one processor, and an image generation model comprising parameters stored in the at least one memory and trained to generate image features for an object depicted in a reference image and to generate a synthetic image depicting the object based on an input prompt and the image features, where the image generation model receives the image features via at least one attention layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** shows an example of an image processing system according to aspects of the present disclosure.

[0009] FIG. **2** shows an example of a method for generating a synthetic image according to aspects of the present disclosure.

[0010] FIG. **3** shows an example of text-to-image generation according to aspects of the present disclosure.

[0011] FIG. **4** shows an example of a method for generating a synthetic image according to aspects of the present disclosure.

[0012] FIG. **5** shows an example of an image processing apparatus according to aspects of the present disclosure.

[0013] FIG. **6** shows an example of a machine learning model according to aspects of the present disclosure.

[0014] FIG. **7** shows an example of an image generation model according to aspects of the present disclosure.

[0015] FIG. **8** shows an example of a U-Net according to aspects of the present disclosure.

[0016] FIG. **9** shows an example of data flow in an image processing system according to aspects of the present disclosure.

[0017] FIG. **10** shows an example of a method for image processing according to aspects of the present disclosure.

[0018] FIG. **11** shows an example of a computing device according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0019] Aspects of the present disclosure relate to image generation using generative machine learning model. Some embodiments of the disclosure relate to an image generation system that accurately generates synthetic images that include the same visual elements from an object depicted in a reference image. In one aspect, an image generation model is trained to generate image features that includes visual information of the object depicted in the reference image. The image features are provided back to the decoding layers of the image generation model and are combined with the features of the text prompt to ensure that the same visual elements of the object from the reference image is accurately generated in the synthetic image along with additional elements described by the text prompt.

[0020] According to some embodiments, the system generates an object mask based on the reference image. In some cases, the system uses a mask generation network to identify one or more visually interesting regions (that aligns with the object described in the text prompt) in the reference image and labels the regions white. The remaining background region is labeled black. In one aspect, the black-and-white object mask is used as an input to the image generation model to generate the image features. In one aspect, the image features include detailed visual information of the object depicted in the reference image. By using an object mask to generate the image features, the image generation model can generate image features having less or no noisy information (e.g., the visual information of the background region of the reference image is excluded).

[0021] In one aspect, the text prompt describes a relation between an object and a background scene, and the reference image depicts the object. In one aspect, the object depicted in the synthetic image includes the same fine visual details from the object depicted in the reference image. For example, when the reference image depicts a beer can with complicated foreign words/characters (as shown in FIG. **6**), the image generation model is able to generate the synthetic image having the same visual details of the beer can (i.e., the same complicated foreign words/characters). In some aspects, the image generation model generates one or more image features based on the reference image, and the one or more image features are added to one or more attention layers in the decoding layers of the U-Net architecture of the image generation model. Accordingly, the image quality of the synthetic image is improved.

[0022] A subfield in image processing relates to image generation. For example, conventional image generation models receive input conditions such as text, image, and color to generate a synthetic image. Some models receive a text prompt that describes a background scene and a reference image that includes an object to generate an output image depicting the object in the background scene. However, these models are unable to accurately generate the visual features of the object. For example, these models generate synthetic images based on the image feature of the reference image during inference time without training the image generation model to adopt the new features from the image feature. As a result, fine details from the reference image cannot be properly transferred and depicted in the generated synthetic image.

[0023] In some cases, some systems use a pre-trained image encoder to generate an image embedding that captures information of the reference image. For example, the image embedding is a high-dimensional vector that encodes semantic information and contents of the reference image. Then, the systems use an image generation model to generate the synthetic image based on the image embedding. However, this technique raises several issues. For example, when encoding the semantic information, the image embedding can lose spatial relationships and fine-grained details presented in the reference image. For example, due to the compact nature of the image embedding, the image embedding might not capture the intricate details of the reference image. As a result, image embeddings are less suitable for tasks such as high-fidelity image synthesis.

[0024] Accordingly, embodiments of the disclosure improve on conventional image generations models by accurately generating synthetic images depicting an object having the same visual content in an object from a reference image. This is achieved using an image generation model that is trained to generate image features of an object from a reference image and combine the image features to the features of the text prompt to generate the synthetic image. Image features include a more accurate information of the visual elements than an image embedding. For example, image features maintain the spatial structure and preserve high levels of detail and resolution of the object depicted in the reference image.

[0025] In some embodiments, the system identifies the object depicted in the reference image using an object mask. For example, the system includes a mask generation network that identifies a visually interesting region (i.e., the object) and generates the object mask indicating the location of the object within the reference image. By generating the one or more image features based on the reference image and the object mask, the one or more image features include information (such as visual features) of the object and not visual features of the background of the reference image. Accordingly, the image features include less or no noisy information about the object, and thus, the image quality of the object depicted in the synthetic image is enhanced.

[0026] An example system of the inventive concept in image processing is provided with reference to FIGS. **1** and **11**. An example application of the inventive concept in image processing is provided with reference to FIGS. **2-3**. Details regarding the architecture of an image processing apparatus are provided with reference to FIGS. **5-8**. An example of a process for image processing is provided with reference to FIGS. **4** and **9**. A description of an example training process is provided with reference to FIG. **10**.

[0027] Embodiments of the present disclosure include systems and methods that improve on conventional image generation models by accurately generating synthetic images depicting an object based on a reference image that includes the object. For example, the synthetic image depicts the object (including the same visual details) from the reference images in a new background scene (e.g., described by a text prompt). For example, the synthetic images maintain the integrity of the object better than the output images generated by the conventional image generation models. In some embodiments, an image generation model is trained to generate image features based on an object depicted in the reference image. By combining the image features to the features representing the text prompt in the corresponding attention layers, the image generation model can preserve the same visual detail of the object from the reference image in the synthetic image.

[0028] In some embodiments, by using the attention layers of the U-Net architecture of the image generation model to generate the image features from the reference image, the model can selectively attend to important regions of the reference image thus reducing computational demand. In some embodiments, the image generation model is trained using the reference image, and the image generation model is able to adopt the new features. Accordingly, fine details from the reference image can be accurately depicted in the generated synthetic image.

Image Processing

[0029] In FIGS. **1-4**, and **9**, a method, apparatus, non-transitory computer readable medium, and system for image processing are described. The method, apparatus, non-transitory computer readable medium, and system include obtaining an input prompt and a reference image depicting an object, generating, using an image generation model, image features for the object based on the reference image, and generating, using the image generation model, a synthetic image depicting the object based on the input prompt and the image features from the reference image.

[0030] In some aspects, the reference image depicts the object in a first scene and the synthetic image depicts the object in a second scene described by the input prompt. Some examples of the method, apparatus, non-transitory computer readable medium, and system further include generating an object mask that indicates the location of the object in the reference image, where the image features are generated based on the object mask. In some aspects, the image features are conditioned based on a diffusion time step.

[0031] Some examples of the method, apparatus, non-transitory computer readable medium, and system further include generating a plurality of layer-specific image features at a plurality of layers of the image generation model, respectively. In some aspects, the image features are generated based on a plurality of reference images. In some aspects, the input prompt comprises a nonce token corresponding to the object. In some aspects, the image generation model is fine-tuned to generate images depicting the object based on the reference image.

[0032] FIG. **1** shows an example of an image processing system according to aspects of the present disclosure. The example shown includes user **100**, user device **105**, image processing apparatus **110**, cloud **115**, and database **120**. Image processing apparatus **110** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **5**.

[0033] Referring to FIG. **1**, user **100** provides a text prompt and a reference image to image processing apparatus **110** via user device **105** and cloud **115**. For example, the text prompt describes an object and a background scene “A beer can among wild flowers.” In some embodiments, the text prompt includes a nonce token that represents the trained object depicted in the reference image. For example, the text prompt may state “A S* among wild flowers.” For example, the nonce token “S*” represents the beer can depicted in the reference image. Additionally, the reference image that user **100** provided depicts the beer can stated by the text prompt. In response, image processing apparatus **110** generates a synthetic image that depicts the beer can from the reference image with a flower background. Image processing apparatus **110** displays the synthetic image and the text response to user **100** via user device **105** and cloud **115**.

[0034] User device **105** may be a personal computer, laptop computer, mainframe computer, palmtop computer, personal assistant, mobile device, or any other suitable processing apparatus. In some examples, user device **105** includes software that incorporates an image processing application. In some examples, the image processing application on user device **105** may include functions of image processing apparatus **110**.

[0035] A user interface may enable user **100** to interact with user device **105**. In some embodiments, the user interface may include an audio device, such as an external speaker system, an external display device such as a display screen, or an input device (e.g., a remote-controlled device interfaced with the user interface directly or through an I/O controller module). In some cases, a user interface may be a graphical user interface (GUI). In some examples, a user interface may be represented in code in which the code is sent to the user device **105** and rendered locally by a browser. The process of using the image processing apparatus **110** is further described with reference to FIG. **2**.

[0036] Image processing apparatus **110** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **5**. According to some aspects, image processing apparatus **110** includes a computer implemented network comprising a machine learning model, text encoder, a mask generation network, and an image generation model. Image processing apparatus **110** further includes a processor unit, a memory unit, an I/O module, and a training component. In some cases, the training component includes a data preparation component. In some embodiments, image processing apparatus **110** further includes a communication interface, user interface components, and a bus as described with reference to FIG. **11**. Additionally, image processing apparatus **110** communicates with user device **105** and database **120** via cloud **115**. Further detail regarding the operation of image processing apparatus **110** is provided with reference to FIG. **2**.

[0037] In some cases, image processing apparatus **110** is implemented on a server. A server provides one or more functions to users linked by way of one or more of the various networks. In some cases, the server includes a single microprocessor board, which includes a microprocessor responsible for controlling aspects of the server. In some cases, a server uses the microprocessor and protocols to exchange data with other devices/users on one or more of the networks via hypertext transfer protocol (HTTP), and simple mail transfer protocol (SMTP), although other protocols such as file transfer protocol (FTP), and simple network management protocol (SNMP) may also be used. In some cases, a server is configured to send and receive hypertext markup language (HTML) formatted files (e.g., for displaying web pages). In various embodiments, a server comprises a general-purpose computing device, a personal computer, a laptop computer, a mainframe computer, a supercomputer, or any other suitable processing apparatus.

[0038] Cloud **115** is a computer network configured to provide on-demand availability of computer system resources, such as data storage and computing power. In some examples, cloud **115** provides resources without active management by the user (e.g., user **100**). The term cloud is sometimes used to describe data centers available to many users over the Internet. Some large cloud networks have functions distributed over multiple locations from central servers. A server is designated an edge server if the server has a direct or close connection to a user. In some cases, cloud **115** is limited to a single organization. In other examples, cloud **115** is available to many organizations. In one example, cloud **115** includes a multi-layer communications network comprising multiple edge routers and core routers. In another example, cloud **115** is based on a local collection of switches in a single physical location.

[0039] According to some aspects, database **120** stores training data (or training set) including a reference image, an input prompt, and a ground-truth image. Database **120** is an organized collection of data. For example, database **120** stores data in a specified format known as a schema. Database **120** may be structured as a single database, a distributed database, multiple distributed databases, or an emergency backup database. In some cases, a database controller may manage data storage and processing in database **120**. In some cases, a user (e.g., user **100**) interacts with the database controller. In other cases, the database controller may operate automatically without user interaction.

[0040] FIG. **2** shows an example of a method **200** for generating a synthetic image according to aspects of the present disclosure. In some examples, these operations are performed by a system including a processor executing a set of codes to control functional elements of an apparatus. Additionally or alternatively, certain processes are performed using special-purpose hardware. Generally, these operations are performed according to the methods and processes described in accordance with aspects of the present disclosure. In some cases, the operations described herein are composed of various substeps, or are performed in conjunction with other operations.

[0041] Referring to FIG. **2**, a user (e.g., the user described with reference to FIG. **1**) provides a text prompt and a reference image to the image processing apparatus (e.g., the image processing apparatus described with reference to FIGS. **1** and **5**). For example, the text prompt describes “A beer can among wild flowers” and the reference image depicts the beer can. The image processing apparatus generates one or more image features based on the reference image using one or more attention layers of an image generation model. The image processing apparatus generates a synthetic image based on the text prompt and the image features. In one aspect, the synthetic image depicts the object (i.e., the beer can) depicted from the reference image and the background described by the text prompt.

[0042] At operation **205**, the system provides a text prompt and a reference image. In some cases, the operations of this step refer to, or may be performed by, a user as described with reference to FIG. **1**. For example, the user provides a text prompt and a reference image depicting a beer can to the image processing apparatus via a user interface provided by the image processing apparatus on a user device. In some cases, the text prompt describes a relationship between the object and a background scene. For example, the text prompt states “A beer can among wild flowers.” In some cases, a nonce token is used to represent the object in the text prompt. For example, the text prompt may state “A S* among wild flowers,” where the “S*” represents the object.

[0043] At operation **210**, the system generates an image feature based on the reference image. In some cases, the operations of this step refer to, or may be performed by, an image processing apparatus as described with reference to FIGS. **1** and **5**. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some embodiments, the system generates an object mask based on the reference image. For example, the object mask indicates the location of the object within the reference image. The object mask is combined with the reference image as input to the image generation model to generate one or more image features. By using an object mask, the image generation model can accurately generate image features containing important information about the object and exclude information about the background region of the reference image.

[0044] At operation **215**, the system generates a synthetic image based on the text prompt and the image feature. In some cases, the operations of this step refer to, or may be performed by, an image processing apparatus as described with reference to FIGS. **1** and **5**. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some cases, the image generation model generates synthetic images based on a noise input and uses the text prompt and image features as guidance to guide the diffusion process of the image generation model. Further detail on the diffusion process is described with reference to FIG. **7**.

[0045] At operation **220**, the system displays the synthetic image. In some cases, the operations of this step refer to, or may be performed by, an image processing apparatus as described with reference to FIGS. **1** and **5**. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some cases, the synthetic image is displayed on a user device via a user interface of the image processing apparatus and cloud.

[0046] FIG. **3** shows an example of text-to-image generation according to aspects of the present disclosure. The example shown includes machine learning model **300**, text prompt **305**, reference image **310**, image generation model **315**, synthetic image **320**, and conventional output image **325**. In some embodiments, machine learning model **300** is implemented in a user interface, where a user can provide inputs such as text prompt **305** and reference image **310** to the user interface to generate synthetic image **320**.

[0047] Referring to FIG. **3**, image generation model **315** receives text prompt **305** and reference image **310** to generate synthetic image **320**. For example, text prompt **305** states “A backpack in a frozen valley" and reference image **310** depicts the visual features of the backpack. In response, image generation model **315** generates synthetic image **320** depicting the same backpack (with fine details) from reference image **310** and the background described by text prompt **305**. Compared to conventional output image **325**, synthetic image **320** accurately depicts detailed features of the backpack from the reference image. For example, the stickers on the backpack in reference image **310** and the sticks on the backpack in synthetic image **320** are the same. Additionally, the backpack straps from reference image **310** and synthetic image **320** are the same. On the contrary, conventional output image **325** does not depict the fine details of the backpack.

[0048] Machine learning model **300** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **6** and **9**. Text prompt **305** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **6**, **7**, and **9**. Reference image **310** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **6** and **9**.

[0049] Image generation model **315** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **5**, **6**, and **9**. Synthetic image **320** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **6** and **9**.

[0050] FIG. **4** shows an example of a method **400** for generating a synthetic image according to aspects of the present disclosure. In some examples, these operations are performed by a system including a processor executing a set of codes to control functional elements of an apparatus. Additionally or alternatively, certain processes are performed using special-purpose hardware. Generally, these operations are performed according to the methods and processes described in accordance with aspects of the present disclosure. In some cases, the operations described herein are composed of various substeps, or are performed in conjunction with other operations.

[0051] At operation **405**, the system obtains an input prompt and a reference image depicting an object. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some cases, for example, an input prompt refers to a text prompt including text or a set of instructions provided as input to a model to elicit an output. In some cases, the text prompt is used as guidance to guide an image generation model. For example, the text prompt includes text that describes the background of the synthetic image to be generated. In some cases, the reference image refers to an image used as data input to a model to guide the image generation model. In some cases, the image depicts an object, an element, a scene, a color, and/or a background.

[0052] At operation **410**, the system generates, using an image generation model, image features for the object based on the reference image. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some embodiments, one or more attention layers of the image generation model receive the reference image to generate one or more image features. In some embodiments, the image generation model generates layer-specific image features at each of the one or more attention layers. Then, each of the layer-specific image features is input to the corresponding attention layer of the image generation model to generate the synthetic image.

[0053] In some cases, image features refer to the characteristics or patterns within the reference image that are used to represent and capture important information. These image features can be easily processed by a machine learning model (e.g., an image generation model). In some cases, image features are represented as vectors or embeddings in a vector space.

[0054] At operation **415**, the system generates, using the image generation model, a synthetic image depicting the object based on the input prompt and the image features from the reference image. In some cases, the operations of this step refer to, or may be performed by, an image generation model as described with reference to FIGS. **3**, **5**, **6**, and **9**. In some aspects, an attention layer in an image generation model focuses on different parts of the input image (e.g., the reference image) when generating or processing the images. The attention layer is able to assign varying levels of importance to different regions of the reference image and enables the image generation model to effectively capture intricate details and complex patterns of the reference image. As a result, the information contained in the image features can be used to generate high-resolution images.

System Architecture

[0055] In FIGS. **5-8**, an apparatus and system for image processing are described. The apparatus and system include at least one processor, at least one memory storing instructions executable by the at least one processor, and an image generation model comprising parameters stored in the at least one memory and trained to generate image features for an object depicted in a reference image and to generate a synthetic image depicting the object based on an input prompt and the image features, where the image generation model receives the image features via at least one attention layer.

[0056] According to some aspects, the image generation model comprises a diffusion U-Net. In some aspects, the image generation model receives the image features at a plurality of attention layers corresponding to a plurality of decoder layers of the diffusion U-Net. In some aspects, the at least one attention layer comprises a self-attention layer.

[0057] Some examples of the apparatus and system further include a mask generation network comprising parameters stored in the at least one memory and configured to generate an object mask that indicates the location of the object in the reference image. Some examples of the apparatus and system further include a text encoder comprising parameters stored in the at least one memory and configured to encode the input prompt.

[0058] FIG. **5** shows an example of an image processing apparatus **500** according to aspects of the present disclosure. The example shown includes image processing apparatus **500**, processor unit **505**, I/O module **510**, memory unit **515**, and training component **535**. In one aspect, memory unit **515** includes text encoder **520**, mask generation network **525**, and image generation model **530**.

[0059] According to some embodiments of the present disclosure, image processing apparatus **500** includes a computer-implemented artificial neural network (ANN). An ANN is a hardware or a software component that includes a number of connected nodes (e.g., artificial neurons), which loosely correspond to the neurons in a human brain. Each connection, or edge, transmits a signal from one node to another (like the physical synapses in a brain). When a node receives a signal, the node processes the signal and then transmits the processed signal to other connected nodes. In some cases, the signals between nodes comprise real numbers, and the output of each node is computed by a function of the sum of its inputs. In some examples, nodes may determine the output using other mathematical algorithms (e.g., selecting the max from the inputs as the output) or any other suitable algorithm for activating the node. Each node and edge is associated with one or more node weights that determine how the signal is processed and transmitted. Image processing apparatus **500** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **1**.

[0060] Processor unit **505** is an intelligent hardware device, (e.g., a general-purpose processing component, a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, processor unit **505** is configured to operate a memory array using a memory controller. In other cases, a memory controller is integrated into the processor. In some cases, processor unit **505** is configured to execute computer-readable instructions stored in a memory to perform various functions. In some embodiments, processor unit **505** includes special-purpose components for modem processing, baseband processing, digital signal processing, or transmission processing. Processor unit **505** is an example of, or includes aspects of, the processor described with reference to FIG. **11**.

[0061] I/O module **510** (e.g., an input/output interface) may include an I/O controller. An I/O controller may manage input and output signals for a device. I/O controller may also manage peripherals not integrated into a device. In some cases, an I/O controller may represent a physical connection or port to an external peripheral. In some cases, an I/O controller may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. In other cases, an I/O controller may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, an I/O controller may be implemented as part of a processor. In some cases, a user may interact with a device via an I/O controller or via hardware components controlled by an I/O controller.

[0062] In some examples, I/O module **510** includes a user interface. A user interface may enable a user to interact with a device. In some embodiments, the user interface may include an audio device, such as an external speaker system, an external display device such as a display screen, or an input device (e.g., a remote-control device interfaced with the user interface directly or through an I/O controller module). In some cases, a user interface may be a graphical user interface (GUI). In some examples, a communication interface operates at the boundary between communicating entities and the channel and may also record and process communications. A communication interface is provided herein to enable a processing system coupled to a transceiver (e.g., a transmitter and/or a receiver). In some examples, the transceiver is configured to transmit (or send) and receive signals for a communications device via an antenna. I/O module **510** is an example of, or includes aspects of, the I/O interface described with reference to FIG. **11**.

[0063] Examples of memory unit **515** include random access memory (RAM), read-only memory (ROM), or a hard disk. Examples of memory unit **515** include solid-state memory and a hard disk drive. In some examples, memory unit **515** is used to store computer-readable, computer-executable software including instructions that, when executed, cause a processor to perform various functions described herein.

[0064] In some cases, memory unit **515** includes, among other things, a basic input/output system (BIOS) that controls basic hardware or software operations such as the interaction with peripheral components or devices. In some cases, a memory controller operates memory cells. For example, the memory controller can include a row decoder, column decoder, or both. In some cases, memory cells within memory unit **515** store information in the form of a logical state.

[0065] In one aspect, memory unit **515** includes text encoder **520**, mask generation network **525**, and image generation model **530**. In one aspect, memory unit **515** includes a machine learning model. Memory unit **515** is an example, of, or includes aspects of, the memory subsystem described with reference to FIG. **11**.

[0066] In some cases, a machine learning model is a computational algorithm, model, or system designed to recognize patterns, make predictions, or perform a specific task (for example, image processing) without being explicitly programmed. According to some aspects, the machine learning model is implemented as software stored in memory unit **515** and executable by processor unit **505**, as firmware, as one or more hardware circuits, or as a combination thereof.

[0067] According to some embodiments of the present disclosure, the machine learning model includes an artificial neural network (ANN), which is a hardware or a software component that includes a number of connected nodes (e.g., artificial neurons), that loosely correspond to the neurons in a human brain. Each connection, or edge, transmits a signal from one node to another (like the physical synapses in a brain). When a node receives a signal, the node processes the signal and then transmits the processed signal to other connected nodes. In some cases, the signals between nodes comprise real numbers, and the output of each node is computed by a function of the sum of its inputs. In some examples, nodes may determine the output using other mathematical algorithms (e.g., selecting the max from the inputs as the output) or any other suitable algorithm for activating the node. Each node and edge is associated with one or more node weights that determine how the signal is processed and transmitted.

[0068] During the training process, the one or more node weights are adjusted to increase the accuracy of the result (e.g., by minimizing a loss function that corresponds in some way to the difference between the current result and the target result). The weight of an edge increases or decreases the strength of the signal transmitted between nodes. In some cases, nodes have a threshold below which a signal is not transmitted at all. In some examples, the nodes are aggregated into layers. Different layers perform different transformations on the corresponding inputs. The initial layer is known as the input layer and the last layer is known as the output layer. In some cases, signals traverse certain layers multiple times.

[0069] According to some embodiments, the machine learning model includes a computer-implemented convolutional neural network (CNN). CNN is a class of neural networks commonly used in computer vision or image classification systems. In some cases, a CNN may enable processing of digital images with minimal pre-processing. A CNN may be characterized by the use of convolutional (or cross-correlational) hidden layers. These layers apply a convolution operation to the input before signaling the result to the next layer. Each convolutional node may process data for a limited field of input (e.g., the receptive field). During a forward pass of the CNN, filters at each layer may be convolved across the input volume, computing the dot product between the filter and the input. During the training process, the filters may be modified so that the filters activate when the filters detect a particular feature within the input.

[0070] In one aspect, the machine learning model includes machine learning parameters. Machine learning parameters, also known as model parameters or weights, are variables that provide behavior and characteristics of the machine learning model. Machine learning parameters can be learned or estimated from training data and are used to make predictions or perform tasks based on learned patterns and relationships in the data.

[0071] Machine learning parameters are adjusted during a training process to minimize a loss function or maximize a performance metric. The goal of the training process is to find optimal values for the parameters that allow the machine learning model to make accurate predictions or perform well on the given task.

[0072] For example, during the training process, an algorithm adjusts machine learning parameters to minimize an error or loss between predicted outputs and actual targets according to optimization techniques like gradient descent, stochastic gradient descent, or other optimization algorithms. Once the machine learning parameters are learned from the training data, the machine learning parameters are used to make predictions on new, unseen data.

[0073] According to some embodiments, the machine learning model includes a computer-implemented recurrent neural network (RNN). An RNN is a class of ANN in which connections between nodes form a directed graph along an ordered (e.g., a temporal) sequence. This enables an RNN to model temporally dynamic behavior such as predicting what element should come next in a sequence. Thus, an RNN is suitable for tasks that involve ordered sequences such as text recognition (where words are ordered in a sentence). In some cases, an RNN includes one or more finite impulse recurrent networks (characterized by nodes forming a directed acyclic graph), one or more infinite impulse recurrent networks (characterized by nodes forming a directed cyclic graph), or a combination thereof.

[0074] According to some embodiments, the machine learning model includes a transformer (or a transformer model, or a transformer network), where the transformer is a type of neural network model used for natural language processing tasks. A transformer network transforms one sequence into another sequence using an encoder and a decoder. The encoder and decoder include modules that can be stacked on top of each other multiple times. The modules comprise multi-head attention and feed-forward layers. The inputs and outputs (target sentences) are first embedded into an n-dimensional space. Positional encoding of the different words (e.g., give each word/part in a sequence a relative position since the sequence depends on the order of its elements) is added to the embedded representation (n-dimensional vector) of each word. In some examples, a transformer network includes an attention mechanism, where the attention looks at an input sequence and decides at each step which other parts of the sequence are important. The attention mechanism involves a query, keys, and values denoted by Q, K, and V, respectively. Q is a matrix that contains the query (vector representation of one word in the sequence), K are the keys (vector representations of the words in the sequence), and V are the values, which are again the vector representations of the words in the sequence. For the encoder and decoder, multi-head attention modules, V consists of the same word sequence as Q. However, for the attention module that takes into account the encoder and the decoder sequences, V is different from the sequence represented by Q. In some cases, values in V are multiplied and summed with some attention-weights a.

[0075] In the machine learning field, an attention mechanism (e.g., implemented in one or more ANNs) is a method of placing differing levels of importance on different elements of an input. Calculating attention may involve three basic steps. First, a similarity between the query and key vectors obtained from the input is computed to generate attention weights. Similarity functions used for this process can include the dot product, splice, detector, and the like. Next, a softmax function is used to normalize the attention weights. Finally, the attention weights are weighed together with the corresponding values. In the context of an attention network, the key and value are vectors or matrices that are used to represent the input data. The key is used to determine which parts of the input the attention mechanism should focus on, while the value is used to represent the actual data being processed.

[0076] An attention mechanism is a key component in some ANN architectures, particularly ANNs employed in natural language processing (NLP) and sequence-to-sequence tasks, which allows an ANN to focus on different parts of an input sequence when making predictions or generating output. Some sequence models (such as RNNs) process an input sequence sequentially, maintaining an internal hidden state that captures information from previous steps. However, in some cases, this sequential processing leads to difficulties in capturing long-range dependencies or attending to specific parts of the input sequence.

[0077] The attention mechanism addresses these difficulties by enabling an ANN to selectively focus on different parts of an input sequence, assigning varying degrees of importance or attention to each part. The attention mechanism achieves the selective focus by considering a relevance of each input element with respect to a current state of the ANN.

[0078] The term “self-attention” refers to a machine learning model in which representations of the input interact with each other to determine attention weights for the input. Self-attention can be distinguished from other attention models because the attention weights are determined at least in part by the input itself.

[0079] According to some aspects, text encoder **520** is a computational algorithm, model, or system designed to convert input text data into a numerical representation in the form of vectors or embeddings. The numerical representation is used for various natural language processing tasks, such as text classification and information retrieval. In some cases, the numerical representation captures meaningful information and relationships within the text in a format that can be easily processed by a machine learning model.

[0080] According to some aspects, text encoder **520** is implemented as software stored in memory unit **515** and executable by processor unit **505**, as firmware, as one or more hardware circuits, or as a combination thereof. According to some aspects, the input prompt includes a nonce token corresponding to the object. According to some aspects, text encoder **520** comprises parameters stored in the at least one memory and configured to encode the input prompt. Text encoder **520** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **7** and **9**.

[0081] According to some aspects, mask generation network **525** is implemented as software stored in memory unit **515** and executable by processor unit **505**, as firmware, as one or more hardware circuits, or as a combination thereof. According to some aspects, mask generation network **525** generates an object mask that indicates a location of the object in the reference image. According to some aspects, mask generation network **525** comprises parameters stored in the at least one memory and configured to generate an object mask that indicates a location of the object in the reference image.

[0082] According to some aspects, image generation model **530** is implemented as software stored in memory unit **515** and executable by processor unit **505**, as firmware, as one or more hardware circuits, or as a combination thereof. According to some aspects, image generation model **530** obtains an input prompt and a reference image depicting an object. In some examples, image generation model **530** generates image features for the object based on the reference image. In some examples, image generation model **530** generates a synthetic image depicting the object based on the input prompt and the image features, where the image generation model **530** is trained to receive the image features via at least one attention layer.

[0083] In some aspects, the synthetic image depicts the object in a scene described by the input prompt. In some aspects, the image features are conditioned based on a diffusion time-step. In some examples, image generation model **530** generates a set of layer-specific image features at a set of layers of the image generation model **530**, respectively, where the set of layer-specific image features are provided to the set of layers as input. In some aspects, the image features are generated based on a set of reference images. In some aspects, the image generation model **530** is trained using the reference image.

[0084] In some aspects, the image generation model **530** is pre-trained in a first training phase without receiving image features at the at least one attention layer and fine-tuned in a second training phase to receive the image features at the at least one attention layer. In some aspects, each layer of the image generation model **530** is updated during the second training phase. In some aspects, the at least one attention layer receives a different number of input tokens during the first training phase and the second training phase. In some aspects, the image generation model **530** is trained to receive layer-specific image features for the object at a set of different layers.

[0085] According to some aspects, image generation model **530** comprises parameters stored in the at least one memory and trained to generate image features for an object depicted in a reference image and to generate a synthetic image depicting the object based on an input prompt and the image features, wherein the image generation model **530** receives the image features via at least one attention layer. In some aspects, the image generation model **530** includes a diffusion U-Net. In some aspects, the image generation model **530** receives the image features at a set of attention layers corresponding to a set of decoder layers of the diffusion U-Net. In some aspects, the at least one attention layer includes a self-attention layer. Image generation model **530** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **6**, and **9**. Image generation model **530** is an example of, or includes aspects of, the diffusion model described with reference to FIG. **7**.

[0086] According to some aspects, training component **535** is implemented as software stored in memory unit **515** and executable by processor unit **505**, as firmware, as one or more hardware circuits, or as a combination thereof. In one aspect, training component **535** includes a data generation component, where the data generation component creates a training set to train the machine learning model.

[0087] According to some embodiments, training component **535** is implemented as software stored in a memory unit and executable by a processor in a processor unit of a separate computing device, as firmware in a separate computing device, as one or more hardware circuits of the separate computing device, or as a combination thereof. In some examples, training component **535** is part of another apparatus other than image processing apparatus **500** and communicates with the image processing apparatus **500**. In some examples, training component **535** is part of image processing apparatus **500**.

[0088] According to some aspects, training component **535** obtains a training set including a reference image depicting an object, an input prompt, and a ground-truth image. In some examples, training component **535** initializes image generation model **530**. In some examples, training component **535** trains, using the training set, the image generation model **530** to generate image features for the object based on the reference image and to generate a synthetic image depicting the object based on the input prompt and the image features, where the image generation model **530** receives the image features via at least one attention layer. In some examples, training component **535** computes a diffusion loss. In some examples, training component **535** updates parameters of the image generation model **530** based on the computed diffusion loss.

[0089] FIG. **6** shows an example of a machine learning model **600** according to aspects of the present disclosure. The example shown includes machine learning model **600**, text prompt **605**, noise input **610**, image generation model **615**, reference image **620**, object mask **625**, image feature **630**, and synthetic image **635**.

[0090] Referring to FIG. **6**, image generation model **615** receives text prompt **605**, noise input **610**, reference image **620**, and object mask **625** to generate synthetic image **635**. In some embodiments, text prompt **605** and reference image **620** are user inputs and noise input **610** and object mask **625** are system-generated inputs obtained from machine learning model **600**. For example, text prompt **605** states "A beer can among wild flowers" and reference image **620** depicts a beer can. In some cases, reference image **620** is used during training to train image generation model **615**. In some embodiments, text prompt **605** may include a nonce token "S*" that indicates the object depicted in reference image **620** learned during the training stage. For example, text prompt **605** may state "A S* among wild flowers."

[0091] In some aspects, image generation model **615** includes a diffusion model (e.g., the diffusion model described with reference to FIG. **7**). Image generation model **615** takes noise input **610** (e.g., a noise map) and iteratively removes noise by performing reverse diffusion to generate synthetic image **635**. In some cases, text prompt **605** is used to guide the diffusion model, where synthetic image **635** includes an element described by text prompt **605**. In some cases, other features such as image feature **630** from reference image **620** can be used for the image generation. Further detail on reverse diffusion is described with reference to FIG. **7**.

[0092] According to some embodiments, image generation model **615** generates an image feature **630** (or one or more layer-specific image features) based on reference image **620** and object mask **625**. For example, a mask generation network is used to generate object mask **625** indicating the location of the object in reference image **620**. By using object mask **625** as input, image generation model **615** can accurately and efficiently generate the image feature **630** for reference image **620**. For example, object mask **625** allows image generation model **615** to attend to the object depicted in reference image **620** rather than other regions (such as the background) of reference image **620**. As a result, less time is needed to generate the image feature **630**, and accurate visual information in image feature **630** can be obtained.

[0093] According to some embodiments, image generation model **615** receives the image feature **630** as input to one or more attention layers of image generation model **615** for the image generation process. For example, the attention layers enable machine learning model **600** (including image generation model **615**) to capture intricate details and complex patterns of reference image **620** and generate synthetic image **635** having the same intricate details and complex patterns. In some cases, an attention layer includes a self-attention layer that routes information from every spatial location in an image (e.g., reference image **620**) to every other. According to some embodiments, the self-attention layer is trained to learn information from spatial locations in the training images (e.g., reference image **620**) to pass information to regions in synthetic image **635** where the desired object is to be generated.

[0094] According to some aspects, image generation model **615** includes a diffusion U-Net architecture, where the U-Net is used to extract the image feature **630** from reference image **620**. In some cases, the image feature **630** is aligned with the generation process before initiating the training process. For each self-attention layer in the decoder (or neural network layers in the up-sampling process) of the U-Net, the image feature **630** from the K and V (key and value representations, respectively) are stored and concatenated during the generation process. Further detail on U-Net is described with reference to FIGS. **7** and **8**.

[0095] In some embodiments, the same diffusion timestep T is used during feature extraction of K and V representations and during image generation. For example, by extracting a different set of K and V values from reference image **620** for each generation timestep T, where T≠0, significant image quality can be improved. In some cases, when timestep Tis set to 0, no noise is applied to reference image **620**. By applying the image features to all layers of the decoder in U-Net, image generation model **615** can generate synthetic image **635** having the highest image quality.

[0096] Machine learning model **600** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **9**. Text prompt **605** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **7**, and **9**. Noise input **610** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **9**.

[0097] Image generation model **615** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **5**, and **9**. Reference image **620** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **9**. Object mask **625** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **9**. Synthetic image **635** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **9**.

[0098] FIG. **7** shows an example of an image generation model according to aspects of the present disclosure. The example shown includes diffusion model **700**, original image **705**, pixel space **710**, image encoder **715**, original image feature **720**, latent space **725**, forward diffusion process **730**, noisy feature **735**, reverse diffusion process **740**, denoised image feature **745**, image decoder **750**, output image **755**, text prompt **760**, text encoder **765**, guidance feature **770**, and guidance space **775**.

[0099] Diffusion models are a class of generative neural networks that can be trained to generate new data with features similar to features found in training data. In particular, diffusion models can be used to generate novel images. Diffusion models can be used for various image generation tasks including image super-resolution, generation of images with perceptual metrics, conditional generation (e.g., generation based on text guidance, color guidance, style guidance, and image guidance), image inpainting, and image manipulation.

[0100] Types of diffusion models include Denoising Diffusion Probabilistic Models (DDPMs) and Denoising Diffusion Implicit Models (DDIMs). In DDPMs, the generative process includes reversing a stochastic Markov diffusion process. DDIMs, on the other hand, use a deterministic process so that the same input results in the same output. Diffusion models may also be characterized by whether the noise is added to the image itself, or to image features generated by an encoder (e.g., latent diffusion).

[0101] Diffusion models work by iteratively adding noise to the data during a forward process and then learning to recover the data by denoising the data during a reverse process. For example, during training, diffusion model **700** may take an original image **705** in a pixel space **710** as input and apply an image encoder **715** to convert original image **705** into original image feature **720** in a latent space **725**. Then, a forward diffusion process **730** gradually adds noise to the original image feature **720** to obtain noisy feature **735** (also in latent space **725**) at various noise levels.

[0102] Next, a reverse diffusion process **740** (e.g., a U-Net ANN) gradually removes the noise from the noisy feature **735** at the various noise levels to obtain the denoised image feature **745** in latent space **725**. In some examples, denoised image feature **745** is compared to the original image feature **720** at each of the various noise levels, and parameters of the reverse diffusion process **740** of the diffusion model are updated based on the comparison. Finally, an image decoder **750** decodes the denoised image feature **745** to obtain an output image **755** in pixel space **710**. In some cases, an output image **755** is generated at each of the various noise levels. The output image **755** can be compared to the original image **705** to train the reverse diffusion process **740**. In some cases, output image **755** refers to the synthetic image (e.g., described with reference to FIGS. **3**, **6**, and **9**).

[0103] In some cases, image encoder **715** and image decoder **750** are pre-trained prior to training the reverse diffusion process **740**. In some examples, image encoder **715** and image decoder **750** are trained jointly, or the image encoder **715** and image decoder **750** are fine-tuned jointly with the reverse diffusion process **740**.

[0104] The reverse diffusion process **740** can also be guided based on a text prompt **760**, or another guidance prompt, such as an image, a layout, a style, a color, a segmentation map, etc. The text prompt **760** can be encoded using a text encoder **765** (e.g., a multimodal encoder) to obtain guidance feature **770** in guidance space **775**. The guidance feature **770** can be combined with the noisy feature **735** at one or more layers of the reverse diffusion process **740** to ensure that the output image **755** includes content described by the text prompt **760**. For example, guidance feature **770** can be combined with the noisy feature **735** using a cross-attention block within the reverse diffusion process **740**. In some cases, text prompt **760** refers to the corresponding element described with reference to FIGS. **3**, **6**, and **9**.

[0105] Cross-attention, also known as multi-head attention, is an extension of the attention mechanism used in some ANNs, for example, for NLP tasks. In some cases, cross-attention attends to multiple parts of an input sequence simultaneously, capturing interactions and dependencies between different elements. In cross-attention, there are two input sequences: a query sequence and a key-value sequence. The query sequence represents the elements that require attention, while the key-value sequence contains the elements to attend to. In some cases, to compute cross-attention, the cross-attention block transforms (for example, using linear projection) each element in the query sequence into a “query” representation, while the elements in the key-value sequence are transformed into “key” and “value” representations.

[0106] The cross-attention block calculates attention scores by measuring the similarity between each query representation and the key representations, where a higher similarity indicates that more attention is given to a key element. An attention score indicates the importance or relevance of each key element to a corresponding query element.

[0107] The cross-attention block then normalizes the attention scores to obtain attention weights (for example, using a softmax function), where the attention weights determine how much information from each value element is incorporated into the final attended representation. By attending to different parts of the key-value sequence simultaneously, the cross-attention block captures relationships and dependencies across the input sequences, allowing the machine learning model to understand the context and generate more accurate and contextually relevant outputs.

[0108] In some examples, diffusion models are based on a neural network architecture known as a U-Net. The U-Net takes input features having an initial resolution and an initial number of channels and processes the input features using an initial neural network layer (e.g., a convolutional network layer) to generate intermediate features. The intermediate features are then down-sampled using a down-sampling layer such that down-sampled features have a resolution less than the initial resolution and a number of channels greater than the initial number of channels.

[0109] This process is repeated multiple times, and then the process is reversed. For example, the down-sampled features are up-sampled using the up-sampling process to obtain up-sampled features. The up-sampled features can be combined with intermediate features having the same resolution and number of channels via a skip connection. These inputs are processed using a final neural network layer to produce output features. In some cases, the output features have the same resolution as the initial resolution and the same number of channels as the initial number of channels.

[0110] In some cases, a U-Net takes additional input features to produce conditionally generated output. For example, the additional input features may include a vector representation of an input prompt. The additional input features can be combined with the intermediate features within the neural network at one or more layers. For example, a cross-attention module can be used to combine the additional input features and the intermediate features.

[0111] A diffusion process may also be modified based on conditional guidance. In some cases, a user provides a text prompt (e.g., text prompt **760**) describing content to be included in a generated image. For example, a user may provide the prompt “I want to generate something creative with this person, with painting style of Van Gough, can you help me?” In some examples, guidance can be provided in a form other than text, such as via an image, a sketch, a color, a style, or a layout. The system converts text prompt **760** (or other guidance) into a conditional guidance vector or other multi-dimensional representation. For example, text may be converted into a vector or a series of vectors using a transformer model, or a multi-modal encoder. In some cases, the encoder for the conditional guidance is trained independently of the diffusion model.

[0112] A noise map is initialized that includes random noise. The noise map may be in a pixel space or a latent space. By initializing an image with random noise, different variations of an image including the content described by the conditional guidance can be generated. Then, the diffusion model **700** generates an image based on the noise map and the conditional guidance vector.

[0113] A diffusion process can include both a forward diffusion process **730** for adding noise to an image (e.g., original image **705**) or features (e.g., original image feature **720**) in a latent space **725** and a reverse diffusion process **740** for denoising the images (or features) to obtain a denoised image (e.g., output image **755**). The forward diffusion process **730** can be represented as q(x.sub.t|x.sub.t−1), and the reverse diffusion process **740** can be represented as p(x.sub.t−1|x.sub.t). In some cases, the forward diffusion process **730** is used during training to generate images with successively greater noise, and a neural network is trained to perform the reverse diffusion process **740** (e.g., to successively remove the noise).

[0114] In an example forward diffusion process **730** for a latent diffusion model (e.g., diffusion model **700**), the diffusion model **700** maps an observed variable x.sub.0 (either in a pixel space **710** or a latent space **725**) intermediate variables x.sub.1, . . . , x.sub.T using a Markov chain. The Markov chain gradually adds Gaussian noise to the data to obtain the approximate posterior q(x.sub.1:T|x.sub.0) as the latent variables are passed through a neural network such as a U-Net, where x.sub.1, . . . , x.sub.T have the same dimensionality as x.sub.0.

[0115] The neural network may be trained to perform the reverse diffusion process **740**. During the reverse diffusion process **740**, the diffusion model **700** begins with noisy data x.sub.T, such as a noisy image and denoises the data to obtain the p(x.sub.t−1|x.sub.t). At each step t−1, the reverse diffusion process **740** takes x.sub.t, such as the first intermediate image, and t as input. Here, t represents a step in the sequence of transitions associated with different noise levels, The reverse diffusion process **740** outputs x.sub.t−1, such as the second intermediate image iteratively until x.sub.T is reverted back to x.sub.0, the original image **705**. The reverse diffusion process **740** can be represented as:

[00001] $p\ (x_{t-1}\ .\mathrm{Math.}\ x_t) := N(x_{t-1};\ \ (x_t, t),\ \ (x_t, t)).$ (1)

[0116] The joint probability of a sequence of samples in the Markov chain can be written as a product of conditionals and the marginal probability:

[00002] $x_T : p\ (x_{0:T}) := p(x_T)\ .\mathrm{Math.}_{t=1}^{T}\ p\ (x_{t-1}\ .\mathrm{Math.}\ x_t),$ (2)

where p(x.sub.T)=N(x.sub.T; 0,I) is the pure noise distribution as the reverse diffusion process **740** takes the outcome of the forward diffusion process **730**, a sample of pure noise, as input and Π.sub.t=1.sup.Tp.sub.θ(x.sub.t−1|x.sub.t) represents a sequence of Gaussian transitions corresponding to a sequence of addition of Gaussian noise to the sample.

[0117] At interference time, observed data x, in a pixel space can be mapped into a latent space **725** as input and a generated data {tilde over (x)} is mapped back into the pixel space **710** from the latent space **725** as output. In some examples, x.sub.0 represents an original input image with low

image quality, latent variables x.sub.1, . . . , x.sub.T represent noisy images, and {tilde over (x)} represents the generated image with high image quality.
[0118] A diffusion model **700** may be trained using both a forward diffusion process **730** and a reverse diffusion process **740**. In one example, the user initializes an untrained model. Initialization can include establishing the architecture of the model and establishing initial values for the model parameters. In some cases, the initialization can include setting hyper-parameters such as the number of layers, the resolution and channels of each layer block, the location of skip connections, and the like.
[0119] The system then adds noise to a training image using a forward diffusion process **730** in N stages. In some cases, the forward diffusion process **730** is a fixed process where Gaussian noise is successively added to an image. In latent diffusion models, the Gaussian noise may be successively added to features (e.g., original image feature **720**) in a latent space **725**.
[0120] At each stage n, starting with stage N, a reverse diffusion process **740** is used to predict the image or image features at stage n−1. For example, the reverse diffusion process **740** can predict the noise that was added by the forward diffusion process **730**, and the predicted noise can be removed from the image to obtain the predicted image. In some cases, an original image **705** is predicted at each stage of the training process.
[0121] The training component (e.g., training component described with reference to FIG. **7**) compares predicted image (or image features) at stage n−1 to an actual image (or image features), such as the image at stage n−1 or the original input image. For example, given observed data x, the diffusion model **700** may be trained to minimize the variational upper bound of the negative log-likelihood −log p.sub.θ(x) of the training data. The training component then updates parameters of the diffusion model **700** based on the comparison. For example, parameters of a U-Net may be updated using gradient descent. Time-dependent parameters of the Gaussian transitions can also be learned. Further detail on U-Net is described with reference to FIG. **8**.
[0122] Text prompt **760** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **6**, and **9**. Text encoder **765** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **5** and **9**.
[0123] FIG. **8** shows an example of a U-Net **800** according to aspects of the present disclosure. The example shown includes U-Net **800**, input feature **805**, initial neural network layer **810**, intermediate feature **815**, down-sampling layer **820**, down-sampled feature **825**, up-sampling process **830**, up-sampled feature **835**, skip connection **840**, final neural network layer **845**, and output feature **850**.
[0124] Referring to FIG. **8**, U-Net **800** depicted is an example of, or includes aspects of, the architecture used within the reverse diffusion process described with reference to FIG. **7**. In some examples, diffusion models are based on a neural network architecture known as a U-Net. The U-Net **800** takes input feature **805** having an initial resolution and an initial number of channels and processes the input feature **805** using an initial neural network layer **810** (e.g., a convolutional network layer) to produce intermediate feature **815**. The intermediate feature **815** is then down-sampled using a down-sampling layer **820** such that the down-sampled feature **825** has a resolution less than the initial resolution and a number of channels greater than the initial number of channels.
[0125] This process is repeated multiple times, and then the process is reversed. In some cases, this reversed process is called a decoding process. For example, the down-sampled feature **825** is up-sampled using up-sampling process **830** to obtain up-sampled feature **835**. In some cases, the down-sampled feature **825** is obtained using an intermediate neural network layer (e.g., a self-attention layer).
[0126] In some aspects, a self-attention layer is a mechanism in neural networks that enables each element in a sequence to attend to a different part of the same sequence, allowing the model to weigh the importance of different elements when processing information. In some cases, the self-attention layer allows the machine learning model to focus on different spatial locations within an

image when processing information. For example, the self-attention layer captures the dependencies and relationships between pixels within the image and captures intricate details and complex patterns of an important region of the image.

[0127] The up-sampled feature **835** can be combined with intermediate feature **815** having the same resolution and number of channels via a skip connection **840**. These inputs are processed using a final neural network layer **845** to produce output feature **850**. In some cases, the output feature **850** has the same resolution as the initial resolution and the same number of channels as the initial number of channels.

[0128] In some cases, U-Net **800** takes additional input features to produce conditionally generated output. For example, the additional input features could include a vector representation of an input prompt. In some cases, for example, the additional input feature includes the one or more image features generated, using the image generation model, based on the reference image. The additional input features can be combined with the intermediate feature **815** within the neural network at one or more layers. For example, a cross-attention module can be used to combine the additional input features and the intermediate feature **815**.

[0129] According to some embodiments, the image generation model (e.g., the image generation model described with reference to FIGS. **3**, **5-7**, and **9**) generates image features based on the reference image and the object mask. In some cases, the image features include a plurality of layer-specific image features. Each of the layer-specific image features is combined with the each of the corresponding decoding layers (e.g., the neural network layers in the up-sampling process **830**) of the U-Net **800**. In some cases, the image features (which include visual information of the object depicted in the reference image) is combined with the features (which includes the information of the text prompt). For example, the two types of features are combined using cross-attention, concatenation, convolution operations, or a combination thereof. Further detail on combining the features is described with reference to FIG. **8**.

[0130] FIG. **9** shows an example of data flow in an image processing system according to aspects of the present disclosure. The example shown includes machine learning model **900**, text prompt **905**, text encoder **910**, text embedding **915**, noise input **920**, image generation model **925**, reference image **930**, object mask **935**, image feature **940**, and synthetic image **945**.

[0131] Referring to FIG. **9**, text prompt **905** and reference image **930** are provided to machine learning model **900** to generate synthetic image **945**. In one aspect, machine learning model **900** includes text encoder **910** and image generation model **925**. According to some embodiments, text encoder **910** receives text prompt **905** to generate text embedding **915**. In some cases, an embedding is a numerical representation of words, sentences, documents, or images in a vector space. The embedding is used to encode semantic meaning, relationships, and context of the words, sentences, documents, or images where the encoding can be processed by a machine learning model. For example, an image embedding captures complex visual features in a high-dimensional vector space. For example, a text embedding (e.g., text embedding**915**) includes semantic relationships between words or tokens in a low-dimensional vector space.

[0132] In some embodiments, reference image **930** and object mask **935** are provided to image generation model **925**, and image generation model **925** generate image feature **940** based on reference image **930** and object mask **935**. For example, object mask **935** is obtained from a mask generation network including a saliency prediction model. In some cases, the saliency prediction model identifies an area of interest and labels the area of interest as white. For example, the area of interest is an object depicted in reference image **930**. In some cases, the object may be described (or partially described) by the text prompt. Then, the saliency prediction model labels the remaining region as black. In one aspect, the object mask **935** is a black-and-white image indicating the object from reference image **930**. In one aspect, object mask **935** identifies a location and/or shape of an object in reference image **930**.

[0133] According to some embodiments, image generation model **925** generates one or more image features based on reference image **930** and object mask **935**. For example, image feature **940** may include layer-specific image features at a plurality of decoding layers (or upsampling process of the U-Net described with reference to FIG. **8**) of image generation model **925**. In some cases, the layer-specific image features of reference image **930** are respectively added to each of the image features of the text embedding **915** and noise input **920** of the corresponding decoding layers of image generation model **925**.

[0134] In some embodiments, image generation model **925** takes text embedding **915**, noise input **920**, and image feature **940** as input. For example, machine learning model **900** provides noise input **920** to image generation model **925**. In one aspect, noise input **920** includes a noise map. In some embodiments, noise input **920** is used to initiate the diffusion process in image generation model **925**. Then, text embedding **915** and image feature **940** are concatenated to one or more attention layers of a U-Net within image generation model **925** to guide the diffusion process. Accordingly, image generation model **925** generates synthetic image **945**. Further detail on U-Net is described with reference to FIG. **8**. Further detail on combining the features is described with reference to FIG. **8**.

[0135] Machine learning model **900** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **6**. Text prompt **905** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **6**, and **7**. Text encoder **910** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **5** and **7**. Noise input **920** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **6**.

[0136] Image generation model **925** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3**, **5**, and **6**. Reference image **930** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **6**. Object mask **935** is an example of, or includes aspects of, the corresponding element described with reference to FIG. **6**. Synthetic image **945** is an example of, or includes aspects of, the corresponding element described with reference to FIGS. **3** and **6**.

Training and Evaluation

[0137] In FIG. **10**, a method, apparatus, non-transitory computer readable medium, and system for image processing are described. The method, apparatus, non-transitory computer readable medium, and system include obtaining a training set including a reference image depicting an object, an input prompt, and a ground-truth image and training, using the training set, the image generation model to generate image features for the object based on the reference image and to generate a synthetic image depicting the object based on the input prompt and the image features.

[0138] In some aspects, the image generation model is pre-trained in a first training phase without receiving the image features at an attention layer and fine-tuned in a second training phase to receive the image features at the attention layer. In some aspects, each layer of the image generation model is updated during the second training phase. In some aspects, the attention layer receives a different number of input tokens during the first training phase and the second training phase.

[0139] Some examples of the method, apparatus, non-transitory computer readable medium, and system further include computing a diffusion loss. Some examples further include updating parameters of the image generation model based on the diffusion loss. In some aspects, the image generation model is trained to receive layer-specific image features for the object at a plurality of different layers.

[0140] FIG. **10** shows an example of a method **1000** for image processing according to aspects of the present disclosure. In some examples, these operations are performed by a system including a processor executing a set of codes to control functional elements of an apparatus. Additionally or alternatively, certain processes are performed using special-purpose hardware. Generally, these operations are performed according to the methods and processes described in accordance with aspects of the present disclosure. In some cases, the operations described herein are composed of various substeps, or are performed in conjunction with other operations.

[0141] At operation **1005**, the system obtains a training set including a reference image depicting an object, an input prompt, and a ground-truth image. In some cases, the operations of this step refer to, or may be performed by, a training component as described with reference to FIG. **5**. In some embodiments, for example, the reference image is used during inference and training. In some cases, the ground-truth image includes an element described by the input prompt and an object from the reference image.

[0142] In some embodiments, the system initializes an image generation model. In some cases, a training component (e.g., the training component described with reference to FIG. **5**.) initializes the image generation model. Initialization can include establishing the architecture of the model and initial values for the model parameters. In some cases, the initialization can include setting hyper-parameters such as the number of layers, the resolution and channels of each layer block, the location of skip connections, and the like.

[0143] At operation **1010**, the system trains, using the training set, the image generation model to generate image features for the object based on the reference image and to generate a synthetic image depicting the object based on the input prompt and the image features, where the image generation model receives the image features via at least one attention layer. In some cases, the operations of this step refer to, or may be performed by, a training component as described with reference to FIG. **5**.

[0144] In some embodiments, during training, all parameters of the U-Net of the image generation model are updated. For example, the training component calculates a diffusion loss based on the synthetic image and the ground-truth image and updates the image generation model based on the diffusion loss. In some embodiments, key and value representation in the attention mechanism of attention layers are concatenated in the corresponding attention layers.

Computing Device

[0145] FIG. **11** shows an example of a computing device **1100** according to aspects of the present disclosure. The example shown includes computing device **1100**, processor **1105**, memory subsystem **1110**, communication interface **1115**, I/O interface **1120**, user interface component **1125**, and channel **1130**.

[0146] In some embodiments, computing device **1100** is an example of, or includes aspects of, the image processing apparatus described with reference to FIGS. **1** and **5**. In some embodiments, computing device **1100** includes processor **1105** that can execute instructions stored in memory subsystem **1110** to obtain an input prompt and a reference image depicting an object, generate image features for the object based on the reference image, and generate a synthetic image depicting the object based on the input prompt and the image features, where an image generation model is trained to receive the image features via at least one attention layer.

[0147] According to some embodiments, processor **1105** includes one or more processors. In some cases, processor **1105** is an intelligent hardware device, (e.g., a general-purpose processing component, a digital signal processor (DSP), a central processing unit (CPU), a graphics processing unit (GPU), a microcontroller, an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA), a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or a combination thereof. In some cases, processor **1105** is configured to operate a memory array using a memory controller. In other cases, a memory controller is integrated into processor **1105**. In some cases, processor **1105** is configured to execute computer-readable instructions stored in a memory to perform various functions. In some embodiments, processor **1105** includes special-purpose components for modem processing, baseband processing, digital signal processing, or transmission processing. Processor **1105** is an example of, or includes aspects of, the processor unit described with reference to FIG. **5**.

[0148] According to some embodiments, memory subsystem **1110** includes one or more memory devices. Examples of a memory device include random access memory (RAM), read-only memory (ROM), or a hard disk. Examples of memory devices include solid-state memory and a hard disk drive. In some examples, memory is used to store computer-readable, computer-executable software including instructions that, when executed, cause a processor to perform various functions described herein. In some cases, the memory contains, among other things, a basic input/output system (BIOS) that controls basic hardware or software operations such as the interaction with peripheral components or devices. In some cases, a memory controller operates memory cells. For example, the memory controller can include a row decoder, column decoder, or both. In some cases, memory cells within a memory store information in the form of a logical state. Memory subsystem **1110** is an example of, or includes aspects of, the memory unit described with reference to FIG. **5**.

[0149] According to some embodiments, communication interface **1115** operates at a boundary between communicating entities (such as computing device **1100**, one or more user devices, a cloud, and one or more databases) and channel **1130** and can record and process communications. In some cases, communication interface **1115** is provided to enable a processing system coupled to a transceiver (e.g., a transmitter and/or a receiver). In some examples, the transceiver is configured to transmit (or send) and receive signals for a communications device via an antenna. In some cases, a bus is used in communication interface **1115**.

[0150] According to some embodiments, I/O interface **1120** is controlled by an I/O controller to manage input and output signals for computing device **1100**. In some cases, I/O interface **1120** manages peripherals not integrated into computing device **1100**. In some cases, I/O interface **1120** represents a physical connection or port to an external peripheral. In some cases, the I/O controller uses an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or other known operating system. In some cases, the I/O controller represents or interacts with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller is implemented as a component of a processor. In some cases, a user interacts with a device via I/O interface **1120** or hardware components controlled by the I/O controller. I/O interface **1120** is an example of, or includes aspects of, the I/O module described with reference to FIG. **5**.

[0151] According to some embodiments, user interface component **1125** enables a user to interact with computing device **1100**. In some cases, user interface component **1125** includes an audio device, such as an external speaker system, an external display device such as a display screen, an input device (e.g., a remote-control device interfaced with a user interface directly or through the I/O controller), or a combination thereof.

[0152] The performance of apparatus, systems, and methods of the present disclosure have been evaluated, and results indicate embodiments of the present disclosure have obtained increased performance over existing technology (e.g., conventional image generation models). Example experiments demonstrate that the image processing apparatus based on the present disclosure outperforms conventional image generation models. Details on the example use cases based on embodiments of the present disclosure are described with reference to FIG. **3**.

[0153] The description and drawings described herein represent example configurations and do not represent all the implementations within the scope of the invention. For example, the operations and steps may be rearranged, combined or otherwise modified. Also, structures and devices may be represented in the form of block diagrams to represent the relationship between components and avoid obscuring the described concepts. Similar components or features may have the same name but may have different reference numbers corresponding to different figures.

[0154] Some modifications to the disclosure may be readily apparent to those skilled in the art, and the principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed

herein.
[0155] The described methods may be implemented or performed by devices that include a general-purpose processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof. A general-purpose processor may be a microprocessor, a conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration). Thus, the functions described herein may be implemented in hardware or software and may be executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored in the form of instructions or code on a computer-readable medium.
[0156] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of code or data. A non-transitory storage medium may be any available medium that can be accessed by a computer. For example, non-transitory computer-readable media can comprise random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), compact disk (CD) or other optical disk storage, magnetic disk storage, or any other non-transitory medium for carrying or storing data or code.
[0157] Also, connecting components may be properly termed computer-readable media. For example, if code or data is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technology such as infrared, radio, or microwave signals, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technology are included in the definition of medium. Combinations of media are also included within the scope of computer-readable media.
[0158] In this disclosure and the following, the word "or" indicates an inclusive list such that, for example, the list of X, Y, or Z means X or Y or Z or XY or XZ or YZ or XYZ. Also the phrase "based on" is not used to represent a closed set of conditions. For example, a step that is described as "based on condition A" may be based on both condition A and condition B. In other words, the phrase "based on" shall be construed to mean "based at least in part on." Also, the words "a" or "an" indicate "at least one."

Claims

1. A method comprising: obtaining a reference image an input prompt describing an image element; identifying an object from the reference image; generating, using an image generation model, image features representing the object based on the reference image; and generating, using the image generation model, a synthetic image depicting the image element and the object based on the input prompt and the image features from the reference image.
2. The method of claim 1, wherein: the reference image depicts the object in a first scene and the synthetic image depicts the object in a second scene described by the input prompt.
3. The method of claim 1, wherein generating the image features comprises: generating an object mask that indicates a location of the object in the reference image, wherein the image features are generated based on the object mask.
4. The method of claim 1, wherein: the input prompt describes a location of the object in the reference image, wherein generating the image features comprises determining the location of the object based on the input prompt, and wherein the image features are generated based on the location of the object.
5. The method of claim 1, wherein generating the image features comprises: generating a plurality

of layer-specific image features at a plurality of layers of the image generation model, respectively.
6. The method of claim 1, wherein: the image features are generated based on a plurality of reference images.
7. The method of claim 1, wherein: the input prompt comprises a nonce token corresponding to the object.
8. The method of claim 1, wherein: the image generation model is fine-tuned to generate images depicting the object based on the reference image.
9. A method of training an image generation model, comprising: obtaining a training set including a reference image depicting an object, an input prompt describing an image element, and a ground-truth image depicting the object and the image element; and training, using the training set, the image generation model to generate image features for the object based on the reference image and to generate a synthetic image depicting the object based on the input prompt and the image features.
10. The method of claim 9, wherein: the image generation model is pre-trained in a first training phase without receiving the image features at an attention layer and fine-tuned in a second training phase to receive the image features at the attention layer.
11. The method of claim 10, wherein: each layer of the image generation model is updated during the second training phase.
12. The method of claim 10, wherein: the attention layer receives a different number of input tokens during the first training phase and the second training phase.
13. The method of claim 9, wherein training the image generation model comprises: computing a diffusion loss; and updating parameters of the image generation model based on the diffusion loss.
14. The method of claim 9, wherein: the image generation model is trained to receive layer-specific image features for the object at a plurality of different layers.
15. An apparatus comprising: at least one processor; at least one memory storing instructions executable by the at least one processor; and an image generation model comprising parameters stored in the at least one memory and trained to generate image features for an object depicted in a reference image and to generate a synthetic image depicting the object based on an input prompt and the image features, wherein the image generation model receives the image features via an attention layer.
16. The apparatus of claim 15, wherein: the image generation model comprises a diffusion U-Net.
17. The apparatus of claim 16, wherein: the image generation model receives the image features at a plurality of attention layers corresponding to a plurality of decoder layers of the diffusion U-Net.
18. The apparatus of claim 15, further comprising: a mask generation network configured to generate an object mask that indicates a location of the object in the reference image.
19. The apparatus of claim 15, further comprising: a text encoder configured to encode the input prompt.
20. The apparatus of claim 15, wherein: the attention layer comprises a self-attention layer.